

Amylase Rate, Graph and Topic 1 Consolidation

I can calculate amylase rate from endpoint time, plot and interpret the pH pattern, evaluate the method, and apply Topic 1 knowledge to exam-style evidence.

RATE 1 TIME

show working

pHRATE GRAPH

interpret pattern

EVALUATE

evidence + judgement

[Open the original interactive lesson](#)

What success looks like

- Calculate rate as 1 divided by time with correct substitution and units, or use the stated scaled-rate convention accurately.
- Plot or interpret pH against rate and identify an evidence-supported optimum without overstating precision.
- Evaluate one strength, one limitation and one justified improvement, then transfer a Topic 1 idea to a fresh question.

Retrieve and reconnect

- Explain why shorter endpoint time means faster amylase activity.
- State one controlled variable from the investigation.
- W7: state what working a CALCULATE answer should show.
- Lead-in: calculate $1 \div 50$ and include s^{-1} .

Two trials reached the endpoint in 50 s and 200 s. Which interpretation should you test against the full dataset?

Think first. Point, say, sign or write your choice. Give one reason.

A. The 200 s trial was faster because its time number is larger.

B. The 50 s trial was faster and should have the larger value when $\text{rate} = 1/\text{time}$.

C. Both rates are identical because both reached an endpoint.

Look closely · what is the evidence?

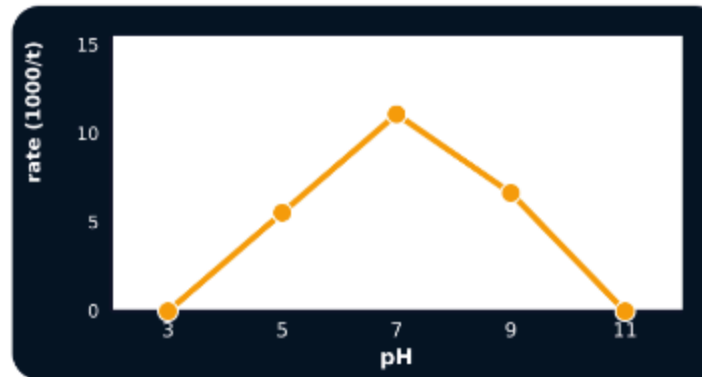
$$\text{rate} = 1 \div \text{time}$$

endpoint 90 s $\rightarrow 1/90 = 0.011$ per second

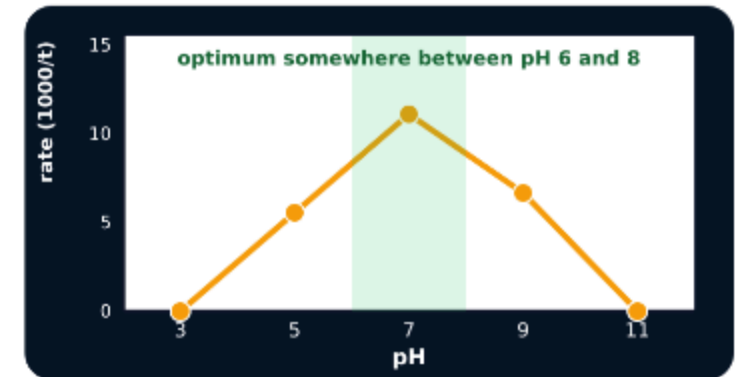
scaled: $1000 \div 90 = 11.1$

SHORT time means a FAST reaction — rate turns that the right way u

Rate equals one divided by time



A pH against rate line graph peaking at pH 7



The same graph with a shaded optimum band

Words we will use

- reciprocal — one divided by
- optimum — highest measured activity
- evaluate — weigh evidence and judge

Watch the model

- Write the relationship before substituting: $\text{rate} = 1/\text{time}$. Keep endpoint time in seconds unless the question states otherwise.
- For 80 s, substitute $\text{rate} = 1/80$ and calculate 0.0125 s^{-1} . Show sufficient working before rounding appropriately.
- Check direction: a shorter time must produce a larger reciprocal rate. If it does not, check the operation and data entry.

Watch the model



Live graph of pH against rate

A trial at pH 6 reaches the endpoint in 125 s. What is its rate to three significant figures?

A. 125 s^{-1}

B. 0.00800 s^{-1}

C. 0.125 s^{-1}

Commit to an answer. Show the evidence you used. Listen, then revise if needed.

Build the explanation

- Plot pH on the independent-variable x-axis and rate on the y-axis. Use an even scale, units and a sensible smooth trend only when the data justify it.
- Identify the highest measured rate as the optimum among the pH values tested. Describe the rise/fall pattern before explaining it with active-site shape and denaturation.
- Evaluate the endpoint method: fixed intervals and controlled variables support comparison; subjective colour judgement and interval-limited timing reduce precision. Justify an improvement such as shorter fixed intervals or an objective colour measure if authorised.

Find and repair the misconception

Graph glitch: join every point sharply, ignore repeats and call the highest plotted pH the exact optimum for all amylase.

A. Keep it: a graph proves an exact universal optimum.

B. Repair it: show the data honestly, use repeats/uncertainty and call it the optimum among the tested conditions.

C. Repair it: remove any value that prevents a smooth curve.

Calculation, graph and evaluation triage

Route each data statement to calculation check, pattern/conclusion or evaluation before writing the full response.

- Choose one evidence statement.
- Commit it to calculation, conclusion or evaluation.
- Repair using the command word and dataset.
- Freeze a chain containing one correct value, one cautious pattern and one justified method judgement.

Calculation, graph and evaluation triage

Use the numbered cards in your pupil pack. Take turns to choose, justify and check. You may direct a partner to move a card.

Choose a group · defend your placement

- CALCULATION CHECK
- PATTERN / CONCLUSION
- EVALUATION

State one result, one explanation and one method judgement, making clear which claim comes from data and which comes from biological knowledge.

Your independent mission

Complete a GCSE-style CP analysis: rate table, pH-rate graph, evidence-led conclusion, method evaluation and a compact Topic 1 transfer check.

Record: Shown rate working, an accurately labelled graph or equivalent data interpretation, a bounded conclusion, a strength-limitation-improvement chain and one Topic 1 transfer response.

Choose your support and challenge

- Supported: Calculate three rates from a worked equation, plot supplied coordinates, choose the tested optimum and complete one limitation-improvement pair.
- Standard: Calculate the full rate column, plot the graph, describe and explain the pattern, then evaluate the endpoint method and answer one Topic 1 exam question.
- Stretch: Analyse repeats and uncertainty, justify the tested optimum, evaluate whether the conclusion is valid and transfer enzyme/transport knowledge to an unfamiliar context.

Show what you know

Use one graph feature to explain how pH affected amylase, then name one justified method improvement.

Point to, read or explain the evidence you made. Choose one next step after the adult has listened.